

# Sunny Zhang

Toronto, ON, Canada

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## Education

### University of Toronto

Honours BSc in Mathematics

*Expected 2029*

GPA: 3.7/4.0

## Technical Skills

**Languages:** Python, C, C++, JavaScript/TypeScript

**Frameworks and Libraries:** React, Flask, FastAPI, NumPy, PyTorch, Matplotlib, Pandas

**Tools:** Git, pytest, CSV, Vercel

## Projects

### Market Regime Forecasting

*Python, Pandas, Flask*

- Implemented a first-order Markov chain to model daily return regimes (down/flat/up) using ~3 months of historical price data (~60 trading days) to estimate next-day state probabilities.
- Cleaned and transformed raw CSV price data into return series; constructed a 3×3 row-normalized transition matrix and validated the probabilistic consistency
- Built a Flask dashboard supporting real-time threshold tuning and 10,000+ Monte Carlo simulations to analyze regime transition sensitivity.
- Wrote unit tests covering state classification and transition logic to ensure transition probability correctness.

### LaunchPilot

*Next.js, TypeScript, FastAPI, Auth0, Backboard.io*

- Built an AI launch copilot at Hack Canada 2026 that turns hackathon projects from idea and GitHub context into market research, positioning, 7-day execution plans, KPIs, and distribution assets.
- Implemented an end-to-end agent workflow across research, positioning, and execution stages, using persistent threads and memory snapshots to preserve project context across the launch pipeline.
- Integrated GitHub context ingestion, Auth0-scoped permissions, and approval-gated execution actions to support secure, repository-aware launch recommendations.
- Shipped with a 2-person team and won Best Use of Backboard.io plus SPUR Build a Real Canadian Startup Top 3/Top 10 recognition.

### UofT Theft Data Map

*Next.js, JavaScript*

- Built an interactive web app visualizing thousands of theft incidents using latitude/longitude coordinates from Toronto Police open data.
- Implemented heatmap rendering and time-of-day aggregation, computing 24-hour risk distributions and hotspot clustering by geographic density.
- Designed a lightweight ETL pipeline to clean and transform raw incident records into map-ready JSON format, reducing data size by ~40% through filtering and normalization.

## Experience

### YM Tech Council

*Sep 2023 – July 2025*

- Supported robotics and programming tournaments by assisting with technical setup, debugging minor software issues, and managing event infrastructure.
- Coordinated scheduling and resource allocation using structured tracking systems and collaborative tools.
- Assisted in organizing student-led technical initiatives involving programming and hardware-based projects.
- Contributed to operational planning for multi-team events requiring real-time coordination and troubleshooting.